REUSABLE MARKET SPECIFICATION

LLC Portfolio Identification and Mailing Research

Process requirements, decision rules, data model, and operating diagrams

---

Chicago / Cook County implementation baseline

Version 1.0 | 28 August 2026

Core rule

Group different LLC owners by the same normalized owner tax mailing address — never by the property address.

1. Purpose and intended use

This specification defines a repeatable process for discovering probable real-estate portfolios held through multiple differently named limited liability companies. It is designed to be reused market by market, beginning with Chicago and Cook County, and to produce a defensible mailing shortlist backed by parcel-level evidence.

The business question

Which owner tax mailing addresses receive records for two or more distinct LLC owners that collectively hold two or more distinct parcels, and what public entity evidence supports the conclusion that those LLCs may share a common controller or operating portfolio?

Required interpretation

- The result is a probable common-controller or common-portfolio signal, not proof of beneficial ownership.
- A manager, registered agent, organizer, attorney, tax processor, or property manager is not automatically the beneficial owner.
- Every mailed candidate must remain traceable to the public source records and the transformations that created the cluster.

Out of scope

- Filtering to multifamily, 2+ unit, commercial, or any other property class before the LLC-owner test.
- Grouping LLCs by the property or situs address.
- Treating a shared registered-agent address as conclusive proof of shared ownership.
- Naming a person as recipient without credible cross-entity support.

2. Executive requirements

ID	Requirement	Mandatory rule
R-01	Universe	Begin with every parcel in the selected geography.
R-02	LLC test	Keep records whose normalized legal owner name ends with LLC.
R-03	Key	Use normalized owner tax mailing address as the candidate-cluster key.
R-04	Threshold	Require 2+ distinct normalized LLC names and 2+ distinct parcels.
R-05	Evidence	Retain property address as inventory evidence only.
R-06	Review	Flag intermediaries and mass-service addresses; never silently remove them.
R-07	Enrichment	Use Secretary of State records after clustering to corroborate entities.
R-08	Outputs	Produce both cluster-level decisions and parcel-level audit evidence.

3. End-to-end workflow

The process separates discovery, qualification, corroboration, and mailing preparation. This prevents entity-enrichment data from biasing the initial portfolio signal.

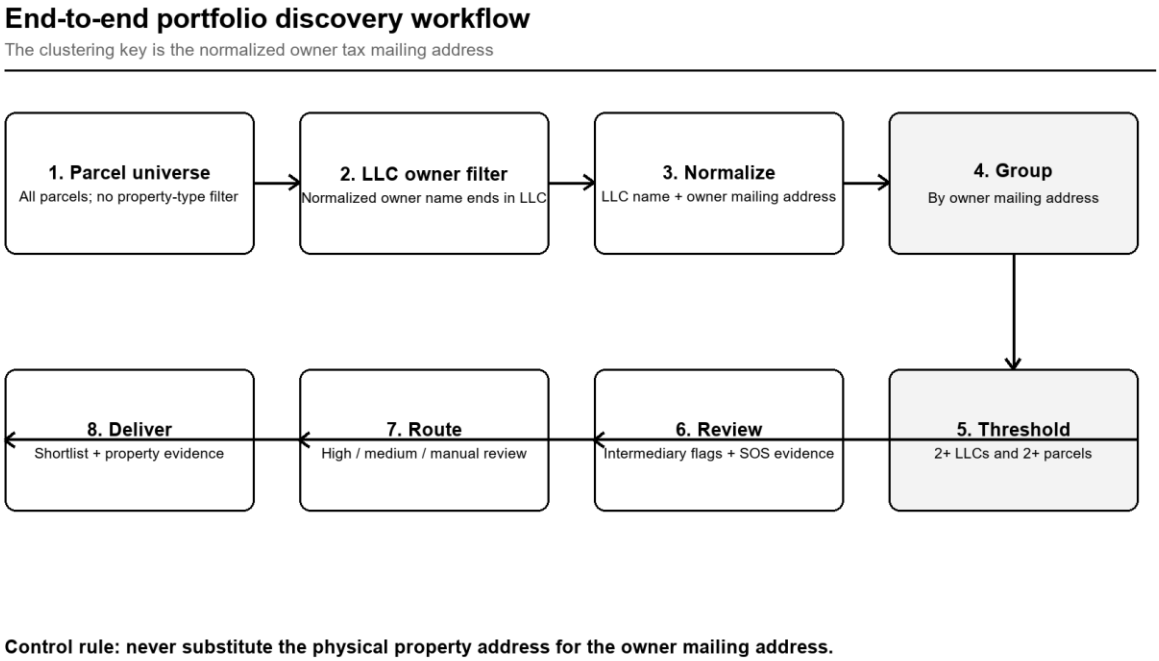


Figure 1. End-to-end portfolio discovery workflow

The owner mailing address is normalized before grouping; Secretary of State evidence enters only after the threshold test.

4. Source data requirements

Minimum parcel-record fields

Field	Meaning	Use	Status
Parcel identifier	Stable parcel/PIN key	Deduplication, evidence, joins	Required
Legal owner name	Tax-roll or assessment owner	LLC suffix filter; entity identity	Required
Owner mailing name	Addressee on tax record	Recipient candidate; QA	Preferred
Owner mailing address	Street, unit, city, state, ZIP	Primary clustering key	Required
Property/situs address	Location of owned parcel	Inventory and evidence only	Required
Source date/version	Assessment roll vintage	Reproducibility	Required

Chicago / Cook County baseline mapping

Cook County field	Role	Processing rule
owner_address_name	Legal owner name	Normalize; test suffix LLC
mail_address_full	Owner tax mailing street/unit	Normalize; cluster key
mail_address_city_name	Mailing city	Cluster-key component
mail_address_state	Mailing state	Cluster-key component
mail_address_zipcode_1	Mailing ZIP	Cluster-key component
prop_address_full	Physical property address	Output/evidence only
pin	Parcel identifier	Distinct parcel count and audit key

5. LLC owner selection

The LLC test applies to the normalized legal owner name. It is intentionally broad and does not depend on property type, unit count, assessed use, or building class.

Required owner-name normalization

- Convert to uppercase and trim leading/trailing whitespace.
- Collapse repeated internal whitespace.
- Standardize terminal variants such as L.L.C., L L C, and punctuation-separated forms to LLC.
- Preserve the original source owner name alongside the normalized value.
- Pass only when the normalized owner name ends with the standalone suffix LLC.

Examples

Source owner name	Normalized owner name	Keep?
Lake View Owner, L.L.C.	LAKE VIEW OWNER LLC	Yes
West Loop Holdings LLC	WEST LOOP HOLDINGS LLC	Yes
LLC Management Services Inc.	LLC MANAGEMENT SERVICES INC	No — does not end in LLC
John and Mary Smith	JOHN AND MARY SMITH	No

6. Correct clustering logic

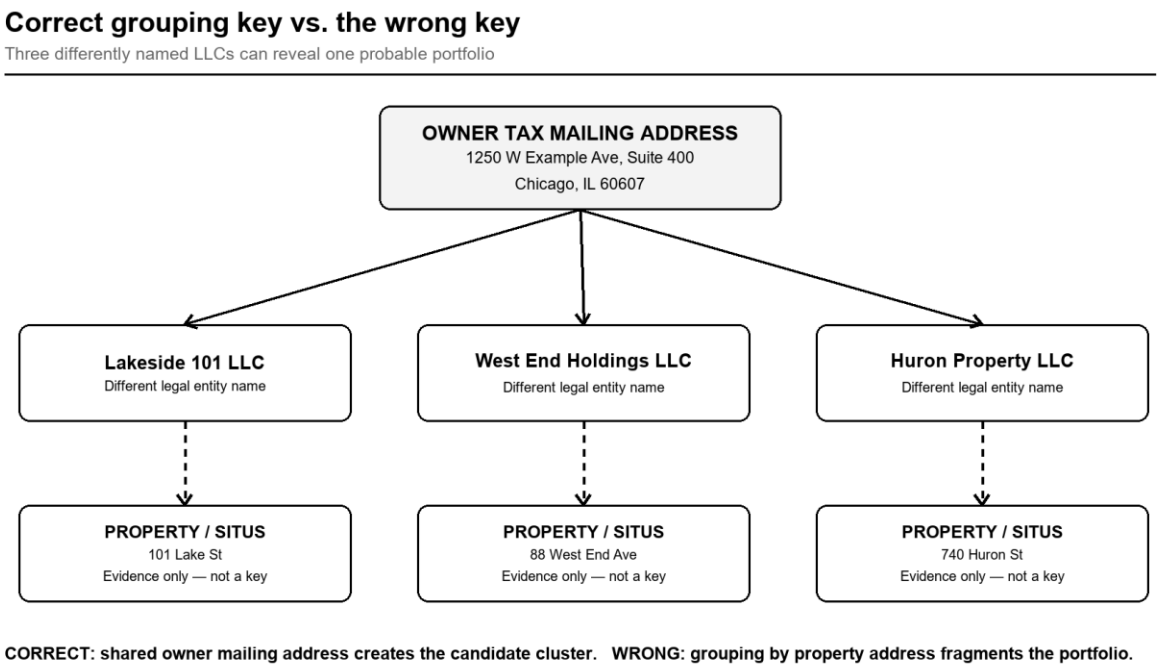


Figure 2. Correct grouping key versus the wrong key

Different property addresses are expected. Their shared owner mailing address is the candidate portfolio signal.

Non-negotiable rule

The physical property address must never be used as the initial grouping key. It identifies the asset, not the place where the owner receives tax mail.

7. Mailing-address normalization

Normalization must improve match consistency without erasing meaningful unit, suite, or postal information. Store both the source value and each normalized component.

Component	Rule
Case and whitespace	Uppercase; trim; collapse repeated spaces
Punctuation	Remove non-semantic punctuation; preserve unit meaning
Street suffixes	Standardize common equivalents, e.g., STREET→ST, AVENUE→AVE
Directionals	Standardize NORTH→N, SOUTH→S, EAST→E, WEST→W when unambiguous
Units	Standardize SUITE/STE and APARTMENT/APT; do not drop the unit
City/state	Normalize spelling and two-letter state abbreviation
ZIP	Use five-digit ZIP as the default; preserve ZIP+4 separately
PO boxes	Standardize PO BOX; retain as a valid mailing address
Missing/invalid	Do not cluster; route to QA/exclusion log

Recommended cluster key

normalized_street_line_1 + normalized_street_line_2 + normalized_city + normalized_state + normalized_zip5

Exact normalized matching is the production baseline. Fuzzy or probabilistic address matching, if later introduced, must be a separate review layer with an explainable score and no silent merges.

8. Data relationship model

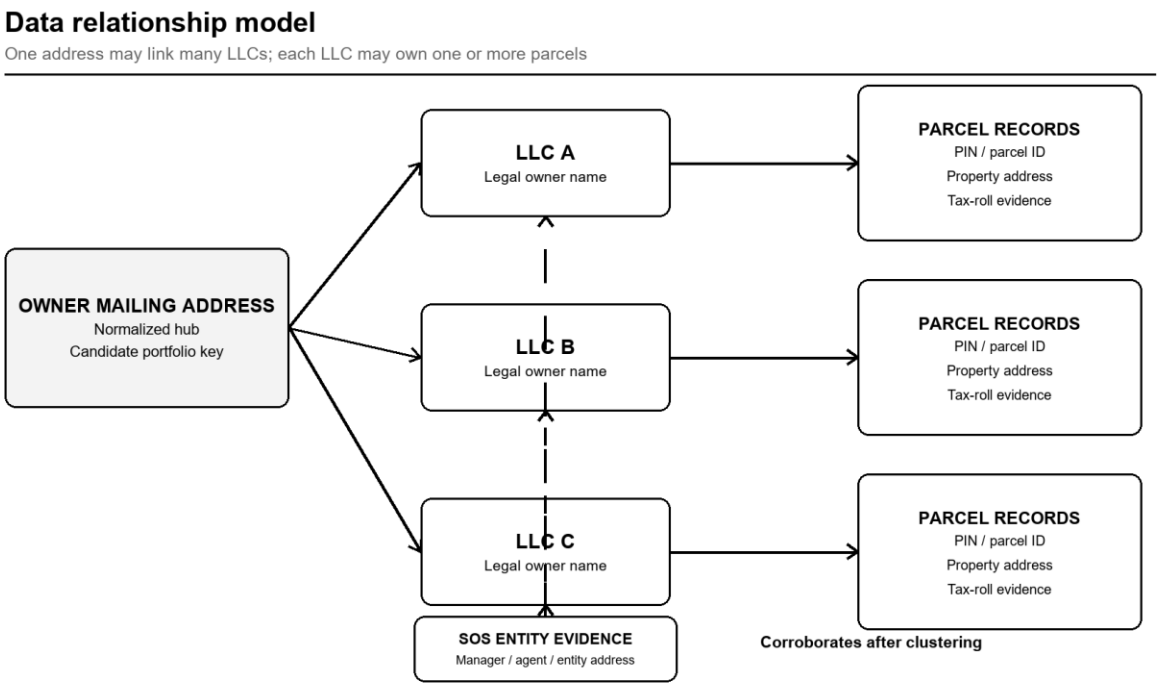


Figure 3. Data relationship model

The address hub creates a candidate cluster; parcel records prove holdings; entity records corroborate after discovery.

9. Cluster construction and qualification

Algorithm

1. Load the complete parcel universe for the target geography and source vintage.
2. Normalize the legal owner name and retain rows where it ends with LLC.
3. Normalize the owner tax mailing address and quarantine missing or invalid values.
4. Group records by the exact normalized owner mailing address.
5. Within each address group, count distinct normalized LLC names and distinct parcel identifiers.
6. Qualify groups with at least two distinct LLC names and at least two distinct parcels.
7. Apply intermediary and mass-service-address flags without deleting the group.
8. Enrich each distinct LLC with Secretary of State evidence and assign confidence.
9. Create the cluster summary, property evidence, mailing shortlist, and QA logs.

Formal acceptance expression

normalized_owner_mailing_address → 2+ distinct normalized LLC names → 2+ distinct parcels

Distinctness and deduplication

- A parcel is distinct by stable parcel/PIN identifier, not by formatted property address.
- An LLC is distinct by normalized legal name within the parcel source; entity identifiers may later refine this.
- Duplicate source rows must not inflate parcel or LLC counts.
- One LLC owning several parcels does not satisfy the multi-LLC requirement by itself.

10. Qualification and routing

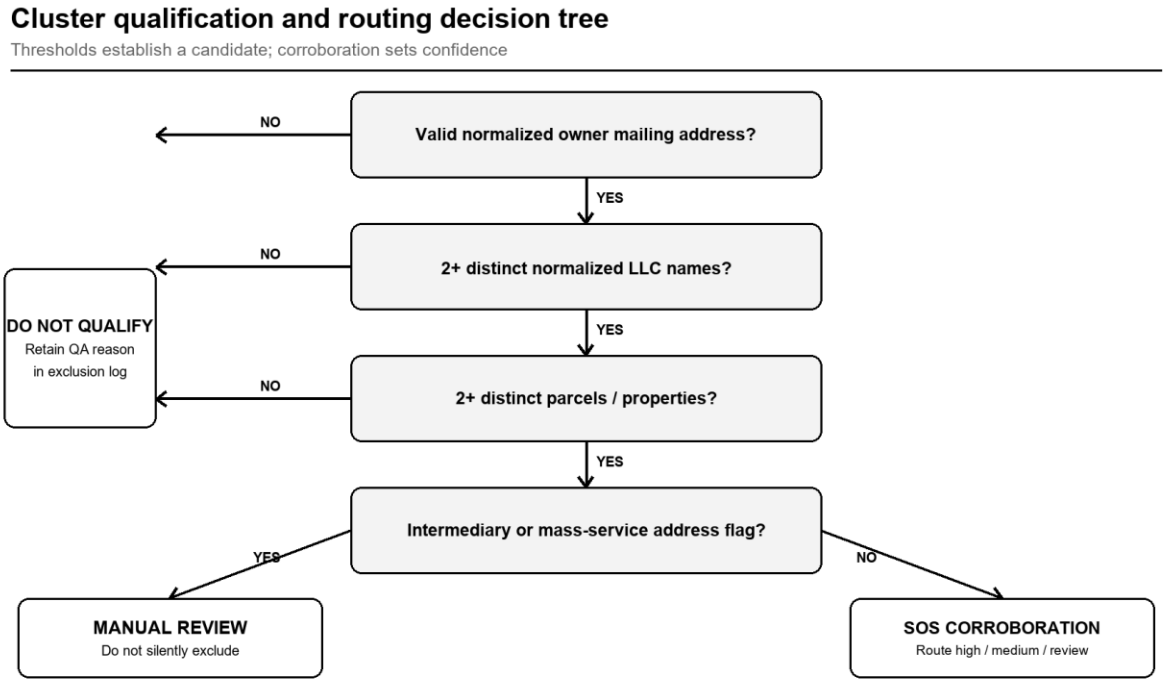


Figure 4. Cluster qualification and routing decision tree

A service-address flag changes the review route; it does not erase the candidate.

11. Intermediary and false-positive controls

Shared addresses may represent a common controller, but they may also belong to professional intermediaries. These cases must be visible, explainable, and reviewable.

Pattern	Observable signal	Required treatment
Registered agent	Address reused across unrelated entities; agent name present	Flag; compare managers/principals and address scale
Law firm	Firm name, attorney suite, many unrelated clients	Flag for manual review
Tax processor/accountant	Tax-service addressee or very high entity volume	Flag; seek corroborating manager evidence
Property manager	Management company receives bills for many owners	Flag; compare owner/entity records
Mass service/virtual office	Address links unusually many LLCs across broad industries	Flag; lower confidence absent corroboration
Property address reused as mail	Mailing address equals one situs address	Flag; do not assume owner residence or control

Scale-based review

The project should calculate the number of distinct LLCs, parcels, owner names, and industry/entity patterns per mailing address. High-volume addresses are not automatically invalid, but require stronger corroboration and often manual review.

12. Secretary of State corroboration

Illinois Secretary of State data is used after parcel clustering. It should validate entity identity and reveal repeated managers, principals, registered agents, or entity addresses across the LLCs in a candidate cluster.

Recommended entity fields

- Entity file number, exact legal name, status, formation date, jurisdiction, and entity address.
- Registered-agent name and address.
- Manager/member or principal names and addresses where publicly available.
- Annual-report or series-LLC relationships where applicable.

Match hierarchy

1. Exact normalized legal name plus jurisdiction/status.
2. Exact legal name plus corroborating address or file number.
3. Fuzzy name candidates only as manual-review leads; never as silent matches.

Interpretation guardrails

- Manager does not necessarily mean beneficial owner.
- Registered-agent matches alone are weak control evidence.
- Repeated credible manager/principal names or addresses across multiple LLCs materially strengthen the portfolio hypothesis.

13. Confidence framework

Level	Evidence pattern	Routing
High	Shared owner mailing address; repeated credible manager/principal or entity-address support across LLCs; no strong intermediary flag	Eligible for mailing shortlist after final QA
Medium	Shared owner mailing address; active entity matches; limited direct manager proof; no strong contradiction	Eligible with conservative recipient naming
Low / review	Agent, law firm, tax processor, property manager, mass-mailing address, situs reuse, ambiguous entity match, or conflicting evidence	Manual review before any mailing

Recipient naming

- Use a person's name only when the same credible person is supported across multiple LLCs in the cluster.
- Otherwise use the tax-roll mailing name when suitable.
- If no defensible recipient name exists, use "Property Owner" rather than inventing a person.
- Retain the LLC names and property list in the evidence file even when the outbound card uses a generalized addressee.

14. Required output package

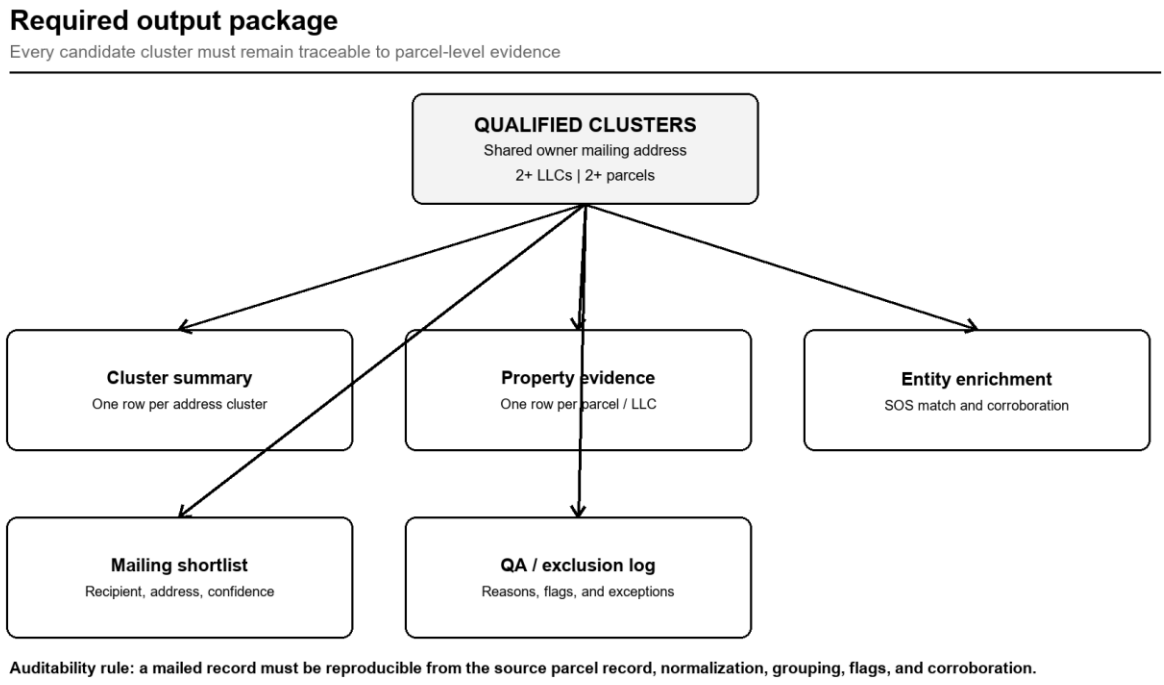


Figure 5. Required output package

The mailing shortlist is a downstream view, not a substitute for cluster and parcel evidence.

15. Output schemas

A. Cluster summary — one row per normalized owner mailing address

Field	Definition
cluster_id	Stable project-generated identifier
normalized_owner_mailing_address	Exact grouping key
source_owner_mailing_addresses	Observed source variants
distinct_llc_count	Count of normalized LLC names
distinct_parcel_count	Count of parcel/PIN values
llc_names	All source and normalized LLC names
property_geography	Cities/neighborhoods/ZIPs represented
intermediary_flags	All triggered review flags
confidence_level	High, medium, or low/review
decision_reason	Human-readable rationale

B. Property evidence — one row per parcel-owner record

Field	Definition
cluster_id	Foreign key to cluster summary
parcel_id / PIN	Stable parcel identifier
source_owner_name	Owner text as published
normalized_llc_name	Normalized distinct-LLC key
property_address	Situs/inventory address; never clustering key
owner_mailing_name	Tax-roll addressee
source_owner_mailing_address	Published mailing address
normalized_owner_mailing_address	Cluster key
source_version	Dataset vintage and retrieval date

C. Entity enrichment and mailing shortlist

Artifact	Minimum fields
Entity enrichment	cluster_id, LLC name, entity/file ID, status, entity address, agent, managers/principals, match method, source date
Mailing shortlist	cluster_id, recipient, deliverable mailing address, LLC count, parcel count, confidence, review status, rationale
QA/exclusion log	record/cluster ID, rule triggered, reason, source values, reviewer, disposition, timestamp

16. Quality assurance and acceptance tests

Record-level tests

- Every included owner name passes the normalized terminal LLC test.
- Every included parcel has a stable parcel/PIN key.
- Every clustered record has a nonblank normalized owner mailing address.
- Every property address is stored separately from the owner mailing address.

Cluster-level tests

- Every qualified cluster contains 2+ distinct normalized LLC names.
- Every qualified cluster contains 2+ distinct parcel identifiers.
- No duplicate source row inflates either count.
- All intermediary flags are retained with the cluster.
- Confidence and recipient decisions have a written rationale.

Auditability tests

- A reviewer can reproduce the group from the normalized address and property evidence file.
- Original source fields are retained beside normalized fields.
- Dataset vintage, retrieval date, and processing version are recorded.
- Manual overrides identify reviewer, timestamp, prior value, new value, and reason.

Acceptance rule

The process is accepted only when 100% of qualified clusters satisfy the two-LLC/two-parcel threshold, 100% of mailed records link to parcel-level evidence, and no property/situs address is used as a clustering key.

17. Market adaptation checklist

1. Define the target geography and obtain the complete parcel universe.
2. Confirm which source fields contain legal owner name, owner mailing name/address, property address, and parcel ID.
3. Document access method, terms, refresh cadence, and dataset vintage.
4. Map local LLC suffix variants and address conventions into the normalization rules.
5. Locate the relevant Secretary of State entity and manager/agent records.
6. Profile high-volume mailing addresses and build local intermediary indicators.
7. Run threshold and QA tests before entity enrichment or mailing selection.
8. Perform a documented sample review across high-, medium-, and review-routed clusters.
9. Freeze the output schema, processing version, and source snapshot for reproducibility.

Market configuration record

Configuration item	Required entry
Market / MSA	Name and geographic boundaries
Parcel authority	Agency, dataset, access method, refresh date
Entity authority	Secretary of State source and refresh date
Field mapping	Local field → canonical field
Normalization version	Named ruleset/version
Thresholds	Default: 2+ LLCs and 2+ parcels
Known intermediaries	Local firms, agents, processors, virtual offices
QA owner	Reviewer and approval date

18. Mailing preparation and operating safeguards

Mailing preparation begins only after cluster qualification, entity review, and recipient routing. The research dataset and the outbound mailing list should remain separate artifacts so later suppression or compliance decisions do not alter the underlying evidence.

- Validate address deliverability and standardization through an authorized postal process before printing.
- Maintain suppression lists for returned mail, opt-outs, do-not-contact decisions, and duplicate households/addresses.
- Do not include sensitive personal information on the card or in exposed tracking parameters.
- Use public-record information only for the documented business purpose and review applicable federal, state, and local solicitation rules before launch.
- Retain the research basis, mailing version, and campaign date so each card can be traced to an approved shortlist record.

Pre-mail release gate

Gate	Required approval
Data	Cluster and parcel evidence passed all acceptance tests
Identity	Recipient name and confidence treatment follow this specification
Address	Deliverability check completed; suppressions applied
Content	Card copy approved; no unsupported ownership claims
Compliance	Campaign reviewed for applicable solicitation/privacy requirements
Archive	Source snapshot, shortlist version, and approval log retained

19. Chicago / Cook County implementation baseline

Chicago is the first implementation market under this reusable specification. Cook County parcel/assessment records provide the owner, mailing, situs, and PIN elements needed for discovery. Illinois Secretary of State entity records provide the post-cluster corroboration layer.

Implementation sequence

1. Acquire and version the countywide parcel source.
2. Apply the LLC suffix test to the legal owner field.
3. Normalize mail_address_full plus city, state, and ZIP into the owner mailing key.
4. Group countywide by that owner mailing key and apply the two-LLC/two-parcel threshold.
5. Profile and flag likely intermediaries and service addresses.
6. Match qualifying LLCs to Illinois entity, agent, annual-report, manager, and series records where available.
7. Review confidence, assign recipient treatment, and publish the five-output package.

Success measure

The Chicago research phase is ready to trigger when the source snapshot, field mapping, normalization rules, intermediary review method, entity-source access, output locations, and QA reviewer are documented and testable against this specification.

20. Definitions and version control

Term	Definition
Owner mailing address	Address where the parcel authority sends owner/tax correspondence; the clustering key
Property/situs address	Physical location of the parcel; inventory evidence only
Candidate cluster	Records sharing one normalized owner mailing address before final confidence review
Qualified cluster	Candidate meeting 2+ LLC and 2+ parcel thresholds
Intermediary flag	Signal that the address may belong to a professional service provider
Corroboration	Independent entity evidence supporting a common-control hypothesis
Probable portfolio	Qualified and reviewed cluster; not a legal conclusion of beneficial ownership

Document control

Version	Date	Status	Change
1.0	28 Aug 2026	Baseline	Initial reusable specification; Chicago/Cook County implementation mapping

Future changes should preserve the core grouping rule unless a formally approved version explicitly replaces it: group different LLC owners by normalized owner tax mailing address, not by property address.